Energy

Types:

- **Potential energy**
  - **Gravitational**
  - Chemical
  - **Elastic/Spring**
  - Nuclear
  - Electric potential (when charge is stored up)
  - Magnetic (when magnets are held apart)
- **Kinetic energy**
  - **Kinetic** (general)
  - Thermal
  - Electrical (when electricity “on” and electrons flowing)
  - Sound
  - Light

Mechanical = total KE and PE of a system (sometimes one or the other)

Energy= the capacity to do work

- **Potential energy–** Energy that is “stored” in some way
  - Ex: chemical energy stored in the bonds of glucose
  - Ex: electrical potential energy stored in a capacitor
  - **Ex: gravitational potential energy stored in an object about to be dropped from a given height**
  - **Ex: spring potential energy = energy stored when a spring or elastic is stretched or compressed**
- **Kinetic energy**– Energy of motion/movement
  - **Kinetic energy (general) = anything moving**
  - **Thermal energy (particles vibrating within something warm or <u>friction</u>**
    - Fun fact, all movement, and thus thermal energy, stops at absolute zero (-273 Kelvin)
  - Ex: electrons moving through your iPhone cable are a form of electromagnetic kinetic energy
- Mechanical = total energy of a system (KE + PE)

Stop here day 1

Energy, Work, and Power

Work-Energy Theorem

"Work"= amount of energy required to do a given task (unit for both energy and work = Joules "J")

- **Work = ΔE** of ANY type!
  - **Basically equal to the <u>change</u> in energy of a given system or "net energy"**
    - **(There must be a *change* for it to count as work)**
  - Ex: if energy "leaves" a system, we say the *system "does work"*
  - Ex: if energy "enters" the system, we say work was done *on the system*
- A lot of times we use energy conservation principles to solve for "work" done when we can't necessarily quantify the energy of a system easily
  - *Ex: using GPE to calculate velocity of a dropped object when we can't directly measure velocity)*

W=Fd

- Another way to calculate work is **Work =Force*distance**
  - ***This means that if F_{net} of a system is 0, Work MUST also be 0***

Energy and Power

"Power" = the rate at which energy is used (when Power is in "Watts", that means a Joule/second)

- Sometimes Power uses another unit of time (ex: "energy per hour", for example) but if not otherwise specified time WILL BE in seconds

Power= Work/time = Δenergy/Δtime = F*d/Δtime

Ex:

a. It takes 2500N to move a car a distance of 5m. What is the work?
- W=Fd = (2500)(5) = **12,500J**

b. If it takes 3 seconds, find the power
- P= W/t = (12500/3)= **4167 watts**

Ex:

Ms. Canty lifts a 50kg weight over her head to a distance of 2m.

a. How much work?
   - W= F*d =
   - 1st: find force→ F=ma= (50kg)($10 m/s^2$) = 500 N
   - (500N)(2m) = 1000 J of work!
b. If it takes her 0.5 seconds, how much power did she apply?
   - P= W/t = 1000/0.5 = 2000 watts

Stop here day 2

Gravitational Potential Energy

$E_{gp} = mgh$

where g is 10 m/s^2 (gravity) and h is height

Makes sense because $W = \Delta E = Fd$

and “mg” represents the force part, “d” is the same as height here!

Stop here day 3

Kinetic energy – energy of motion (in Joules)

$$E_k = \frac{1}{2}mv^2$$

m= mass in kg

v= velocity in m/s

(velocity is same as speed in this context, because if we are solving for it we won't know its sign b/c of the square)

Spring/Elastic potential $E_{elastic}$ or E_{spring} = ½ kx^2

k= spring constant (will be given in Newtons per meter)
x = distance stretched or compressed in meters

Ex: a spring with constant “k” of 140 N/m is compressed 0.05m. How much spring potential energy?
$= \frac{1}{2}(140)(0.05^2) = 0.175J$

Calculating Spring Constant from Force and distance

k = F/d

(force in N, distance in meters!)

Ex: "A spring requires 10,000 N to be compressed 3.0cm"

k = 10,000N /0.03m

= 333,333 N/m

Conservation of Energy

- Energy cannot be created or destroyed, it can only change forms
  - **AKA mechanical energy (total energy of a system) never changes!!!!!!**

- When solving conservation problems, you will set one form of energy equal to the other
  - Ex: GPE= KE

$$mgh = \frac{1}{2} mv^2$$

Problem-Solving Strategies

- What kind of energy do I have to “start” with? (pick one)
  - $E_{elastic}$
  - $E_{gravity}$ /GPE
  - $E_{kinetic}$
  - $E_{thermal}$
- What kind of energy do we “end” with? (^pick one)

THEN

- “Plug in” the equation for each of your energies
- Plug in numbers and solve!

$$\boxed{Energy_{start}} = \boxed{Energy_{end}}$$

Ex:

$$\boxed{E_{gravity}} = \boxed{E_{kinetic}}$$

$$\boxed{mgh} = \boxed{\tfrac{1}{2} mv^2}$$

Ex:

A 25.0kg object is dropped from a height of 10m. What is the maximum velocity it can have at the bottom of the hill?

GPE to KE

$mgh= \frac{1}{2} mv^2$

$\cancel{(25)}(10)(10) = \frac{1}{2} \cancel{(25)}v^2$

$$V^2 = \frac{(100)}{(\frac{1}{2})} = 200$$

v= 14.1 m/s

Name: ______________________ Period: ________

Conservation of Energy Practice

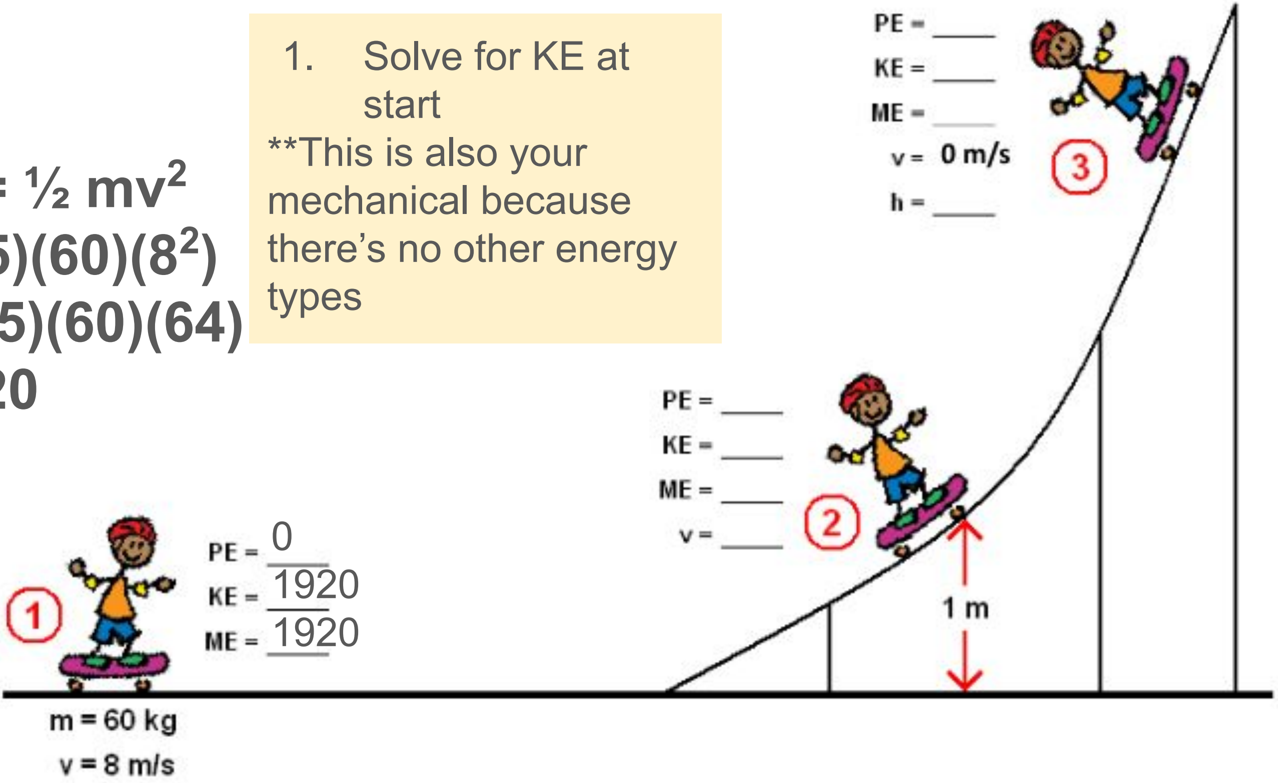
$$KE = \frac{1}{2}mv^2 \qquad v = \sqrt{\frac{2KE}{m}} \qquad PE = mgh \qquad h = \frac{PE}{mg} \qquad ME = KE + PE$$

1. Calculate the potential energy, kinetic energy, mechanical energy, velocity, and height of the skater at the various locations.

1. Solve for KE at start

**This is also your mechanical because there's no other energy types

$KE = ½ mv^2$
$=(0.5)(60)(8^2)$
$= (0.5)(60)(64)$
$=1920$



Name: ______________________ Period: ________

Conservation of Energy Practice

$$KE = \frac{1}{2}mv^2 \qquad v = \sqrt{\frac{2KE}{m}} \qquad PE = mgh \qquad h = \frac{PE}{mg} \qquad ME = KE + PE$$

1. Calculate the potential energy, kinetic energy, mechanical energy, velocity, and height of the skater at the various locations.

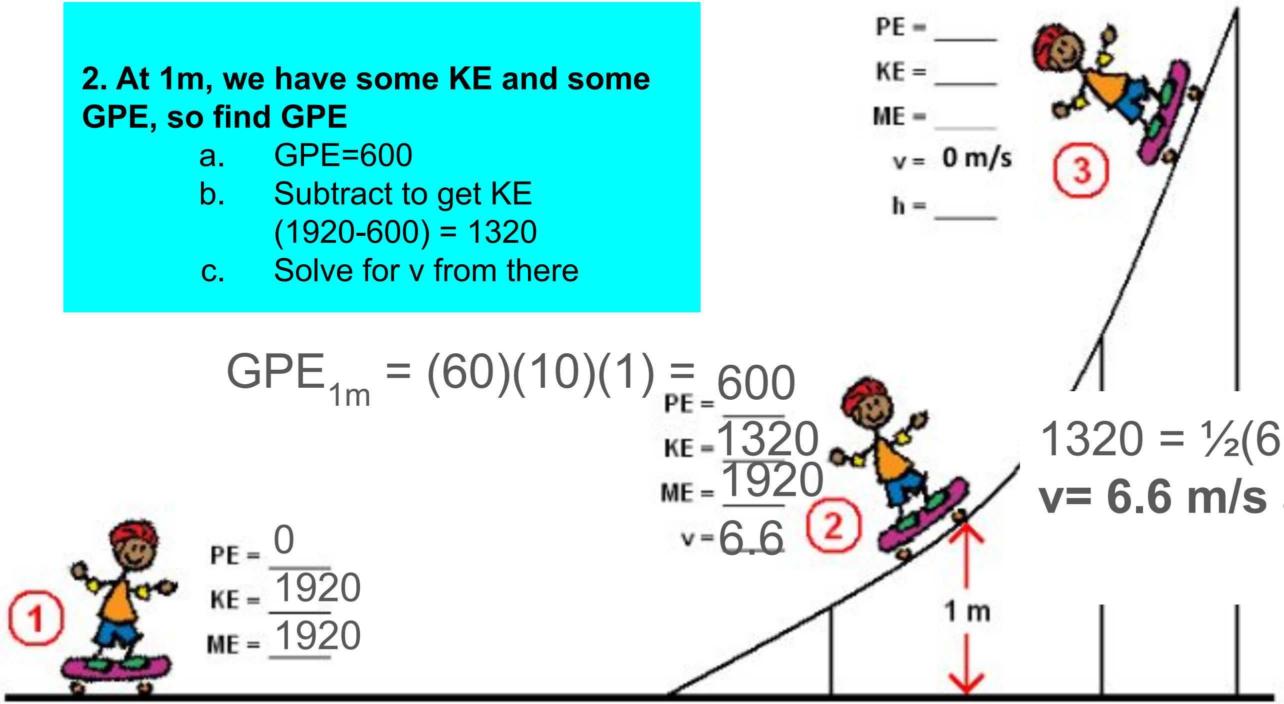
2. At 1m, we have some KE and some GPE, so find GPE

a. GPE=600

b. Subtract to get KE (1920-600) = 1320

c. Solve for v from there

$GPE_{1m} = (60)(10)(1) = 600$



$1320 = \frac{1}{2}(60)v^2$

v= 6.6 m/s at height of 1m

m = 60 kg

v = 8 m/s

Name: ______________________ Period: ________

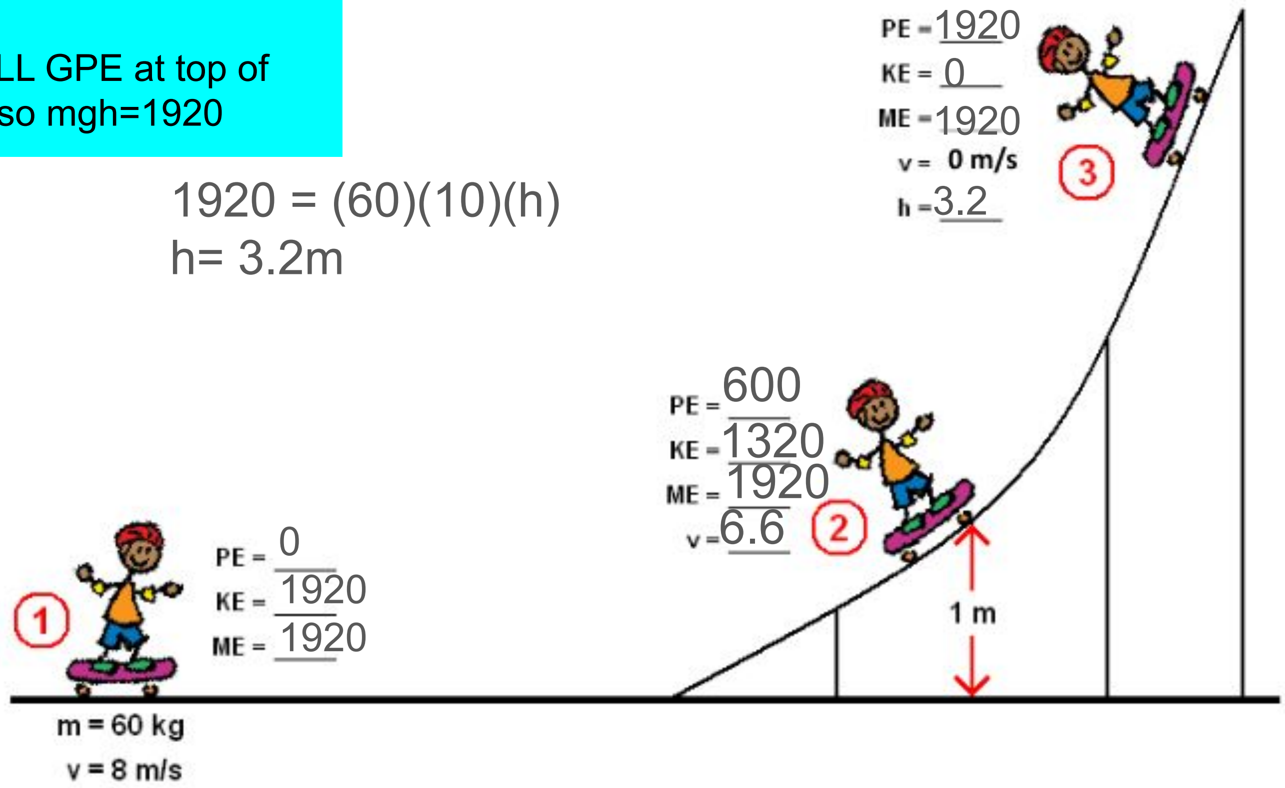
Conservation of Energy Practice

$$KE = \frac{1}{2}mv^2 \qquad v = \sqrt{\frac{2KE}{m}} \qquad PE = mgh \qquad h = \frac{PE}{mg} \qquad ME = KE + PE$$

1. Calculate the potential energy, kinetic energy, mechanical energy, velocity, and height of the skater at the various locations.

3. ALL GPE at top of hill, so mgh=1920

1920 = (60)(10)(h)
h= 3.2m



Stop Here Conservation of Energy

Gravitational potential energy **$GPE = mgh$**

Kinetic energy **$KE = \frac{1}{2}mv^2$**

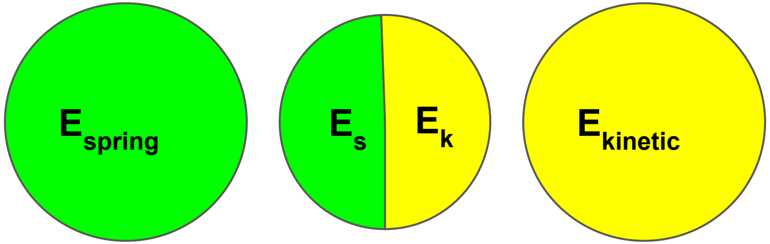
Spring/Elastic potential **$E_{elastic}$ or $E_{spring} = \frac{1}{2}$**
kx^2

m= mass in kg
g= 10 m/s^2 (constant)
h= height in meters
v= velocity in m/s
k= spring constant (will be given)

Stop here 2026

Examples:

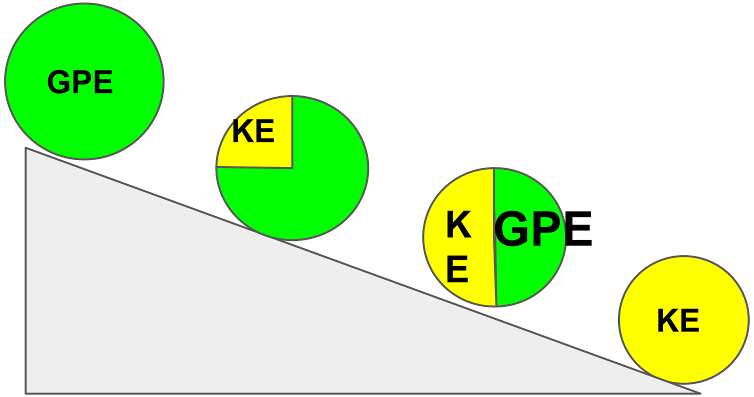
"A wind-up toy is wound up then "walks" across a table



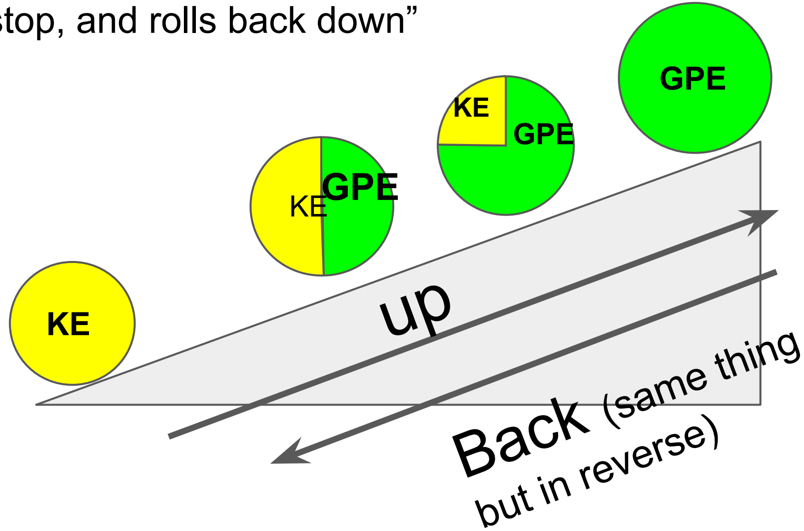
^Es from spring being compressed!

Examples:

“A ball starts at rest and rolls down an incline”



“A ball is rolled up a ramp, comes to a stop, and rolls back down”

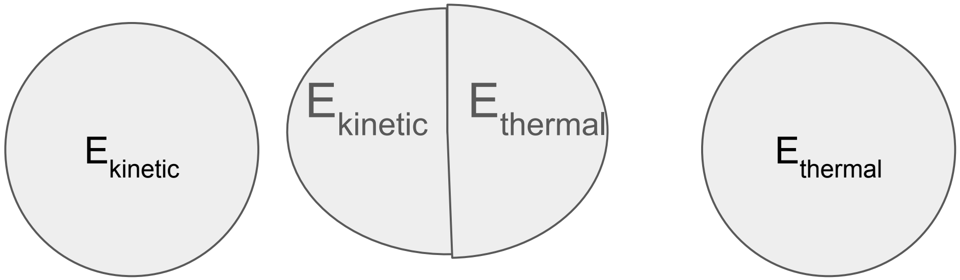


When friction is involved…

We call this "thermal energy" (a form of KE)

You should indicate this on your pie charts when you include friction's effect on an object

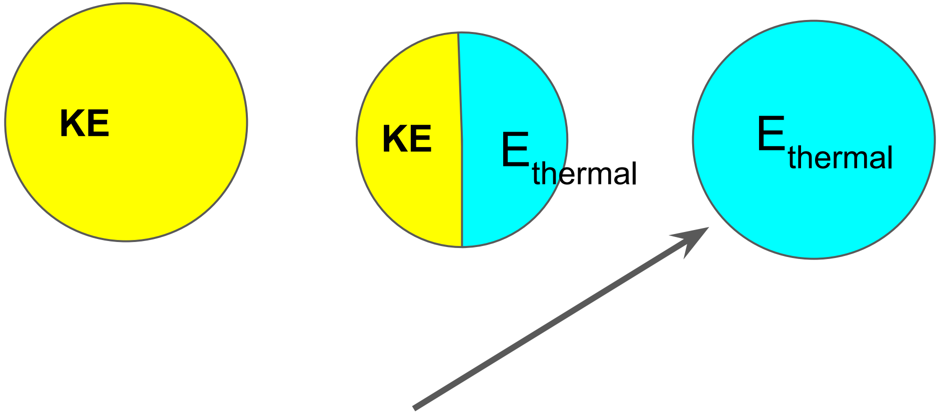
Ex: anything stopping when it hits a surface has its energy dissipated as heat and therefore thermal energy (the "slap the chicken to cook it" hypothetical)



Examples:

“A ball rolls to a stop on a flat surface”

****If the ball is on a flat surface, the only way it could be slowing down is if it was losing energy to friction (transformed to thermal energy!) SO, we have to say that in this case the energy becomes thermal even if we can’t “see” it!*****

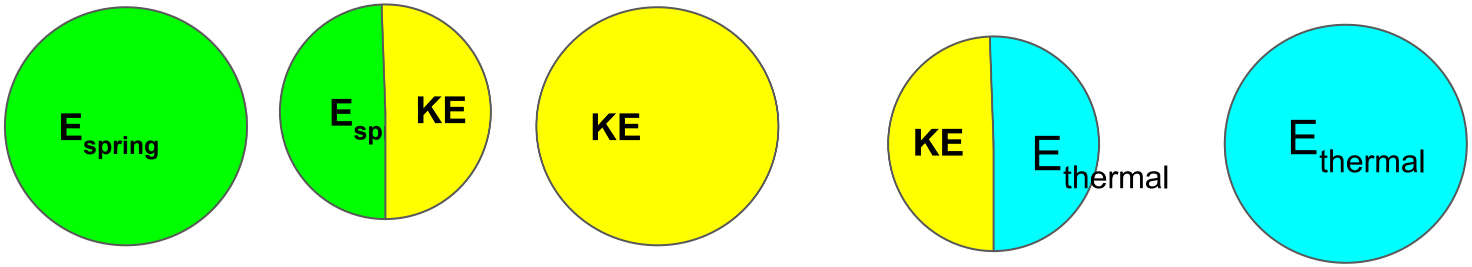


Since thermal energy is a ***type of*** KE; you ***can*** write that it’s kinetic but ***<u>I would prefer thermal</u>*** when you know friction is present because it’s calculated in its own specific way

Examples:

“A wind-up toy is wound up then “walks” across a table, slowing down to a stop”

****If the wind-up toy is on a flat surface, the only way it could be slowing down is if it was losing energy to friction (transformed to thermal energy!) SO, we have to say that in this case the energy becomes thermal even if we can’t “see” it!*****



Since thermal energy is a ***type of*** KE; you ***can*** write that it’s kinetic but I would prefer thermal when you know friction is present

Examples:

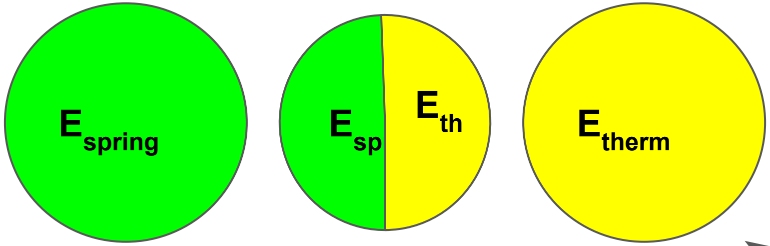
"A wind-up toy is wound up then "walks" across a table, slowing down to a stop"

****If the wind-up toy is on a flat surface, the only way it could be slowing down is if it was losing energy to friction (transformed to thermal energy!) SO, we have to say that in this case the energy becomes thermal even if we can't "see" it!*****

Windup Toy

Demonstration

When you wind up the windup toy, potential energy is stored in its spring. As you wind the spring tighter and tighter, the spring holds more potential energy. The kind of potential energy in the spring of a windup toy is elastic potential energy. Elastic potential energy is stored when something "stretchy" or "springy" is stretched or compressed. When you wind up the windup toy, you compress its spring. When you let go, the spring unwinds, and the toy moves.



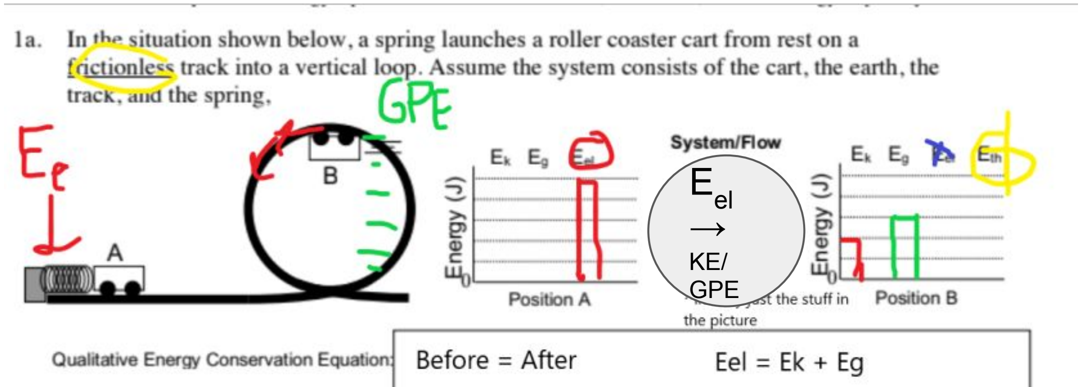
Since thermal energy is a ***type of*** KE; you ***can*** write that it's kinetic but I would prefer thermal when you know friction is present

Stop Day 1 2024

Using Bar Graphs in Energy Conservation

Problems where all components are treated as part of the system:

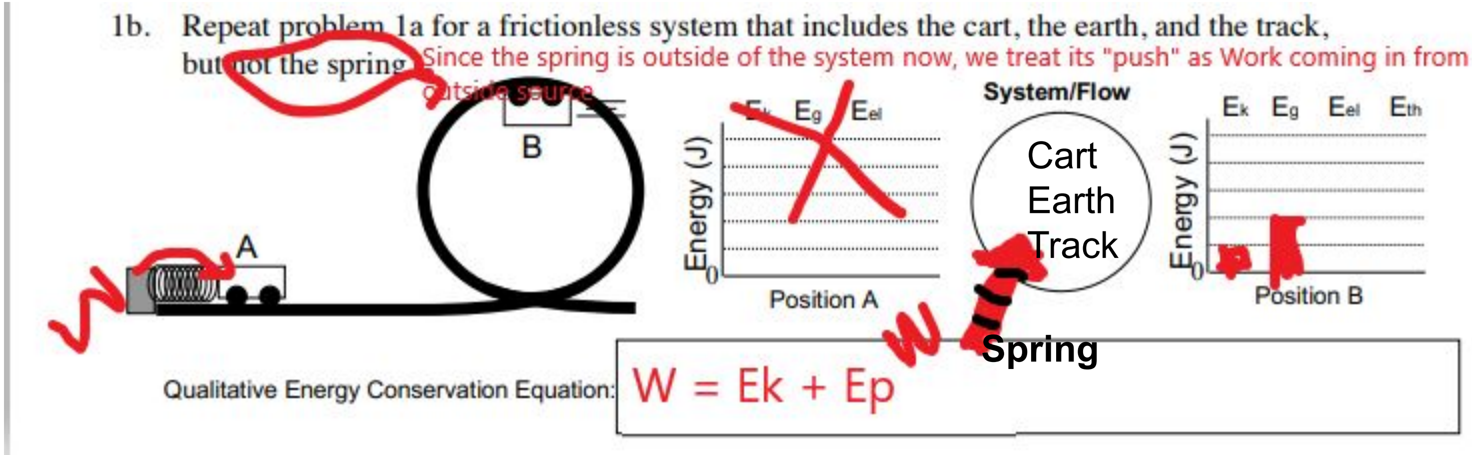
- Anything in “the system” (everything unless otherwise stated) goes on the first bar graph as an energy type, ex: Eg
- The start and end energies MUST add up to be equal because energy is conserved
- The example below is under the assumption that the cart has both KE and GPE at the top, but it wouldn’t be wrong if you only had GPE



Using Bar Graphs in Energy Conservation

Problems with an “outside source” of energy

- Anything in “the system” (everything unless otherwise stated) goes on the first bar graph as an energy type, ex: Eg
- **Anything outside “the system”** (will be specifically stated) **is treated as outside Work (W)** and should be shown as an arrow bar instead of on the graph because it is coming from an outside source
- Ex: “The system does not include the spring”



Calculations

Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Definitions Review:

- **Kinetic Energy**– energy of motion (E_k) or (KE)
  - **Mechanical energy**– generally what we think of as kinetic energy; comes from an object in motion (E_{mech}) or (KE_{mech})*
  - **Thermal energy**– energy contained in a system that is responsible for its temperature (E_{Th}) or (KE_{Th})*
- **Potential Energy**– energy that is stored and can be released (E_p) or (PE)*
  - **Gravitational Potential Energy**– the energy stored in an object that is raised at any height above the ground (or whatever "surface") (E_{gpe}) or (PE_g)*
  - **Chemical Energy**– energy stored in chemical bonds (E_c) or (PE_c)*
  - **Elastic Energy**– energy stored in an item that is stretched or compressed, ex: spring or elastic band (E_{el}) or (PE_{el})
  - **Nuclear Energy**– energy stored within the subatomic particles of the nucleus (E_{nuc}) or (PE_{nuc})
- **Work–** the amount of force necessary to move an object across a given distance (W)

All of the above quantities are measured in "Joules" (J)

Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Equations:

- **Kinetic Energy**
  - **Mechanical energy**– generally what we think of as kinetic energy; comes from an object in motion (E_k) or (KE_{mech})*
    - $KE_{mech} = \frac{1}{2} mv^2$

      where m=mass in kg; v= velocity in m/s

All of the above quantities are measured in "Joules" (J)

Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Calculations:

- **Potential Energy**
  - **Gravitational Potential Energy**– the energy stored in an object that is raised at any height above the ground (or whatever "surface") (E_{gpe}) or (PE_g)*
    - GPE= mgh

      Where m=mass   h= height in meters   g= 9.8 m/s^2 (gravitational constant)

Ex: a ball of mass 0.5kg is raised to a height of 50m.

a. How much GPE does it have?

   GPE= mgh ⇒ (0.5)(10)(50) = <u>250J of GPE</u>

b. How much force did it take to raise the ball to that height?
   i. **W = GPE** because the work it took to raise the ball to that height is conserved, just like energy is!
   ii. Therefore, GPE=W=Fx, so   250J = F(50)   ⇒ <u>F= 5N of force</u>

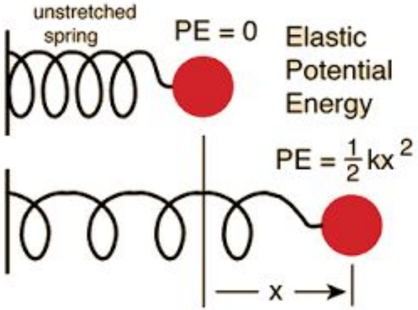
Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Calculations:

- **Potential Energy**
  - **Elastic Energy**– energy stored in an item that is stretched or compressed, ex: spring or elastic band (E_{el}) or (E_s)
    - $E_{el} = \frac{1}{2} kx^2$

      Where x is the distance the item is compressed or stretched; k is the spring constant of that given material (different materials have different "k")



Ex: A spring with a proportionality constant of 20 N/m is stretched over a distance of 0.5m. How much PE is stored in the spring?

$$E_{el} = \frac{1}{2} kx^2 \quad \Rightarrow \quad E_{el} = (\frac{1}{2})(20)(0.5^2) \quad = \quad 2.5 \text{ J}$$

Not given "k"??
You can find it by taking:
#Newtons
#meters
(you'll be given this like "*it takes # newtons to compress a spring X meters*")

All of the above quantities are measured in "Joules" (J)

Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Equations:

- **Thermal energy**– energy contained in a system that is responsible for its temperature (KE_{Th}) or (Q–from chemistry; "enthalpy")*
  - $Q = mc\Delta T$

    where m=mass in kg; c= specific heat capacity; ΔT= change in temperature (in Celcius); Q is KE_{therm} (just another name/variable)

  - $\Delta T = T_{final} - T_{initial}$

All of the above quantities are measured in "Joules" (J)

Stop Here Quant. Energy Calcs.

Energy Equations Wrap-Up :)

Kinetic energy = “KE” = E_k **= $\frac{1}{2} mv^2$**

Gravitational potential energy = “GPE” = E_g = **mgh**

Spring (elastic) potential energy → E_s or E_{el} **= $\frac{1}{2} k(x^2)$** **(E_{el} = synonym for E_s)**

1. Work-energy theorem

   Work = F*d where d represents distance

   Work = ΔE

2. Power, energy, and work

   Power = work/time **P = W/Δt**

3. How force is related to energy :) – If you combine equations 1 and 2, you can see how force is related to Power!

$$P = \frac{\Delta E}{\Delta t} = \frac{F*d}{\Delta t}$$

unit = Watts (1 Watt= 1 J/s = “joule per second”)

Quantitative Energy Calculations

What "variable" you use, Ex: E_k , doesn't necessarily matter as long as YOU know what you mean

Calculations:

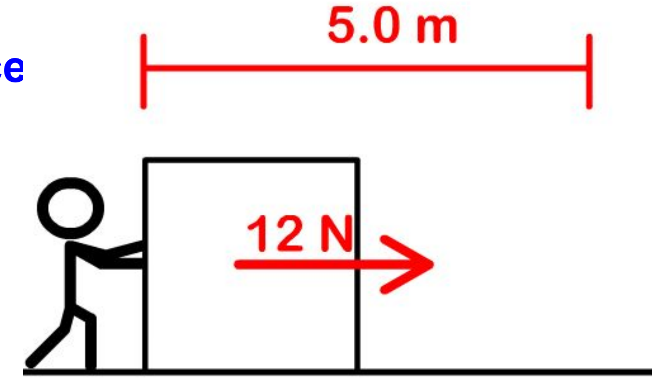
- Work
  - **W= Fd or W=Fx**

    where d or x is the distance the item travels and F is the force applied over that distance

Ex: a person pushes a box across 5m with 12N of force

How much work was done?

W= Fx W= (12)(5) = 60 J of work



All of the above quantities are measured in "Joules" (J)

$1\ kWh = \mathbf{3{,}600{,}000\ J}$

Because 1000J = 1 kWatt, so a kiloWattHour is 1000 x 60 mins x 60 seconds = 3,600,000 J
Fun fact I guess?

Determining Power

- Power = the amount of work done over a given period of time

Formula

$$P = \frac{W}{\Delta t}$$

P = power
W = work
Δt = elapsed time

$$\Delta T = T_{final} - T_{initial}$$

J	**Joules**	Units
s	**Seconds**	←

1 J/s is equal to **1 watt**; we use both tbh

- ***You may be asked to find power without being given "Work"*-- this should be calculated as the first step when it isn't already "given"**

Remember:

Work= force * distance